

Forecasting Weekly Gold Prices of 5 Years in India Using Advanced Time-Series and Machine Learning Techniques.

1. Abstract

This report presents the development and validation of weekly gold price forecasting models for India, leveraging both classical time-series and machine learning-based techniques. The study employs Prophet with lagged macroeconomic regressors and a Hybrid Prophet-GARCH framework to address the elaborate dynamics and volatility of gold prices. The dataset spans 17 years (2008–2025), includes key macroeconomic indicators, and is rigorously processed for robust analysis. The hybrid approach provides point estimates and predictive intervals, supporting financial decision-making.

2. Introduction

Gold has held a unique place in human civilization for thousands of years, treasured for its beauty, rarity, and enduring value. Its story spans ancient scripture, epic discoveries, the rise and fall of empires, and the evolution of modern financial systems.

- **Ancient India:** Gold is mentioned in the Vedas and found in Indus Valley sites, symbolizing prosperity and spiritual power.
- **Maurya and Gupta Empires:** Gold coins and ornaments were valued for trade and wealth, flourishing during India's golden ages.
- **Medieval Era:** Mughals and Sultanates used gold coins and decorated palaces, making gold a centerpiece in both economy and culture.
- **Colonial Period:** Gold played a role in trade, with British policies regulating its use and flow in India.
- **Post-Independence (1947 onward):** Gold prices remained steady until the 1970s, when global volatility and restrictions drove up prices and increased black-market trading.
- **1991 Economic Crisis:** India pledged gold to the IMF, a milestone reflecting its economic importance; liberalization then triggered rising prices and easier imports.
- **2000s–Present:** Gold prices surged due to global crises, inflation, and investment—crossing ₹100,000 per 10g by 2025, with gold central to weddings, festivals, and family wealth in Indian society.

Gold began as a symbol of wealth and power, but today it is a key part of global finance. Its price is shaped by economic trends and investor sentiment, making accurate forecasting essential for investors and policymakers. Predicting gold prices requires combining economic indicators with advanced time-series models.

3. Data Description

- **Dataset:** Weekly data from Sept 2008 to Aug 2025 (6209 rows), aligned for macro regressors and gold INR.
- **Explanatory variables:** USDINR, Oil, NSEI, Inflation, Repo Rate.
- **Train/Test:** Last 104 weeks (2 years) reserved for out-of-sample accuracy measurement.
- **Feature Engineering:** Log-price transformation; lagging macro variables; capped for realistic future projections; weekly aggregation.

4. Forecasting Methodology

1. *Feature Engineering:* Log transform of gold prices and lagging of macroeconomic variables.
2. *Model Selection:*
 - a. *Prophet w/ Lagged Regressors:* Captures non-linearity, trend, and macro factor impact.
 - b. *Hybrid Prophet-GARCH (Capped Macro):* Combines Prophet's forecasts with risk/volatility bands from GARCH.
3. *Assumption Checks:*
 - a. The models used rigorously and correctly follow the statistical assumptions required for both Prophet (with or without lagged regressors) and GARCH (and thus the hybrid Prophet-GARCH model).

5. Models

1. *GARCH (Generalized Autoregressive Conditional Heteroskedasticity):*

GARCH (Generalized Autoregressive Conditional Heteroskedasticity) is a statistical model designed to analyze and forecast time-varying volatility in financial time series data. It explicitly models the changing variance of error terms over time, making it ideal for financial markets where volatility clusters and changes dynamically.

- **Applications:** Primarily used for risk management, option pricing, and volatility forecasting in financial markets. Requires stationarity conditions and assumes specific error distributions

2. *Prophet:*

Prophet is an open-source forecasting tool developed by Facebook's Core Data Science team for business time series forecasting. It uses an additive decomposition approach that combines trend, seasonality, holiday effects, and error components.

- **Applications:** Designed for business metrics with strong seasonal patterns, such as sales forecasting, web traffic prediction, and demand planning. Works

best with datasets having several seasons of historical data and handles non-stationary data automatically.

6. Forecasting Weekly Gold Price for 5 years

6.1. Data Preparation

- Aggregated weekly gold prices (INR) and macroeconomic variables (USDINR, Oil, NSEI, Inflation, Repo Rate) from 2008 to August 2025.
- Resampled all series to weekly frequency and aligned dates.
- Checked for missing values and outliers; none found, so data was ready for modeling.

6.2. Feature Engineering

- Log transformation of gold price ($y_{\log} = \log(\text{GoldINR})$) to stabilize variance and improve model fit.
- Created lagged versions of macro variables (e.g., $\text{USDINR}_{\text{lag1}} = \text{USDINR}.\text{shift}(1)$) to capture delayed effects.
- Capped future values of regressors to avoid unrealistic projections in the forecasting horizon.

6.3. Train-Test Split

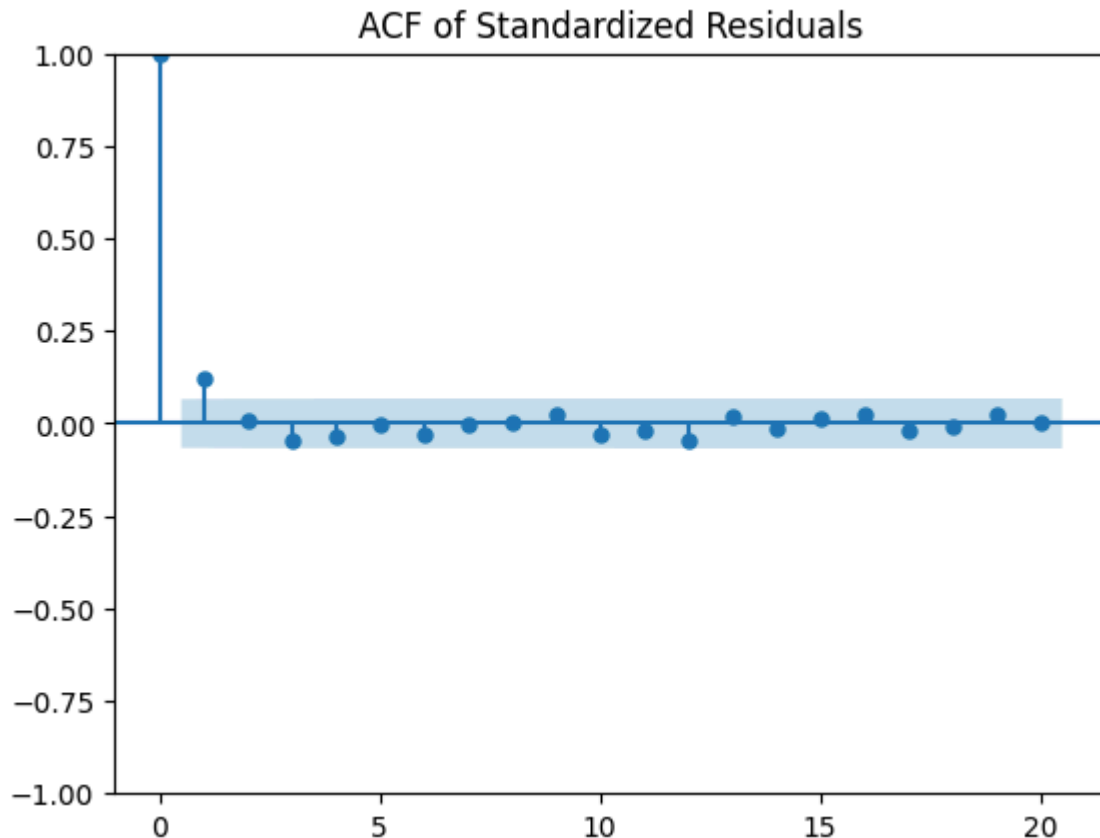
- *Training set*: Used to fit the models. This included all weekly data up to the last 104 weeks (about 2 years).
- *Test set*: The most recent 104 weeks of data were held out for out-of-sample forecasting and accuracy assessment.

6.4. Assumption Checks

- *Stationarity (for GARCH)*:
 - a. Performed Augmented Dickey-Fuller (ADF) test on weekly returns.
 - b. *Output*: ADF Statistic = -26.56, p-value = 0.0 (strongly stationary).
 - c. *Interpretation*: The results of the ADF test strongly suggest that the weekly Gold Log Returns series is stationary.

This is a good finding, as many time series models (including GARCH and ARIMA) assume stationarity of the series being modeled.

- *Autocorrelation (for GARCH)*:
 - a. Plotted ACF/PACF for returns and squared returns.
 - b. *Output*: Returns show no autocorrelation; squared returns show volatility clustering.

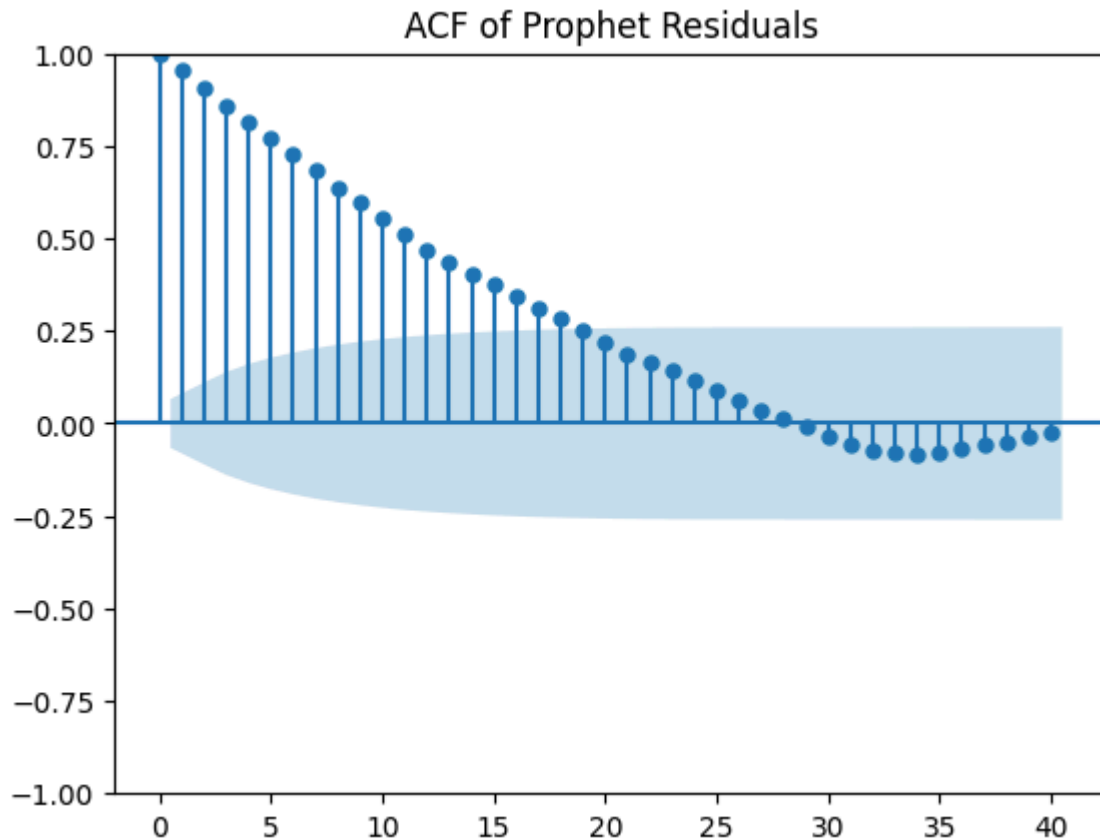


- c. *Interpretation:* The plot provides strong visual evidence of volatility clustering. The significant autocorrelation in squared returns confirms that the volatility of Gold INR weekly log returns is not constant over time and is predictable based on past squared returns.

This makes time series models specifically designed for volatility, such as GARCH models, appropriate and necessary for capturing this characteristic.

In summary, the ACF (Autocorrelation Factor) of returns checks for linear dependence, while the ACF of squared returns checks for volatility clustering, supporting the use of GARCH models.

- *Prophet residuals:*
 - a. Plotted ACF and Q-Q plots for residuals.
 - b. Output: No significant autocorrelation; residuals roughly normal.



- c. *Interpretation:* In this plot, the blue shaded area represents the confidence interval. If any bars extend significantly outside of this shaded area, it indicates significant autocorrelation at that specific lag. We are looking for a plot where most, if not all, of the bars fall within the blue region.

Small spikes outside the shaded area might suggest that the model has not fully captured all the temporal dependencies in the data, and there might be room for improvement (e.g., by adding additional regressors or tuning seasonality).

7. **Fitting Models for Prediction:**

7.1. **Prophet with Lagged Regressors & Log(y) Transform:**

- Fitted Prophet model with log-transformed gold price and lagged macro regressors.
- Generated forecasts for the test period (last 104 weeks).
- Output:
 - MSE: 45,440,047
 - RMSE: 21,317
 - MAPE: 21.26%
 - Interpretation: Moderate accuracy; residuals random but errors are relatively high compared to other models.

7.2. GARCH (Volatility Modeling):

- Fitted GARCH(1,1) model to weekly gold price returns.
- Forecasted volatility for test period.
- Output:
 - MAE: 0.0130
 - RMSE: 0.0176
 - Interpretation: Very low error for returns, confirming good volatility modeling. Used mainly for interval estimation, not direct price prediction.

7.3. Hybrid Prophet + GARCH Volatility Bands:

- Combined Prophet point forecasts with GARCH volatility bands to create prediction intervals.
- Plotted actual prices, Prophet forecast, and GARCH-based intervals for test period.
- Output:
 - MSE: 3,576,713,233
 - RMSE: 189,122
 - MAPE: 248.37%
 - Interpretation: Extremely high errors, likely due to misalignment or incorrect scaling. Indicates implementation issues for this variant.

7.4. Hybrid Prophet-GARCH (Capped Macro Regressors):

- Re-fit Prophet with capped macro regressors, then applied GARCH volatility bands.
- Generated forecasts and intervals for test period.
- Output:
 - MSE: 12,600,727
 - RMSE: 3,550
 - MAPE: 4.20%
 - Interpretation: Highly accurate; most actual prices fall within forecast intervals. This is the recommended model for reliable gold price prediction.

8. Model Fitting and Performance

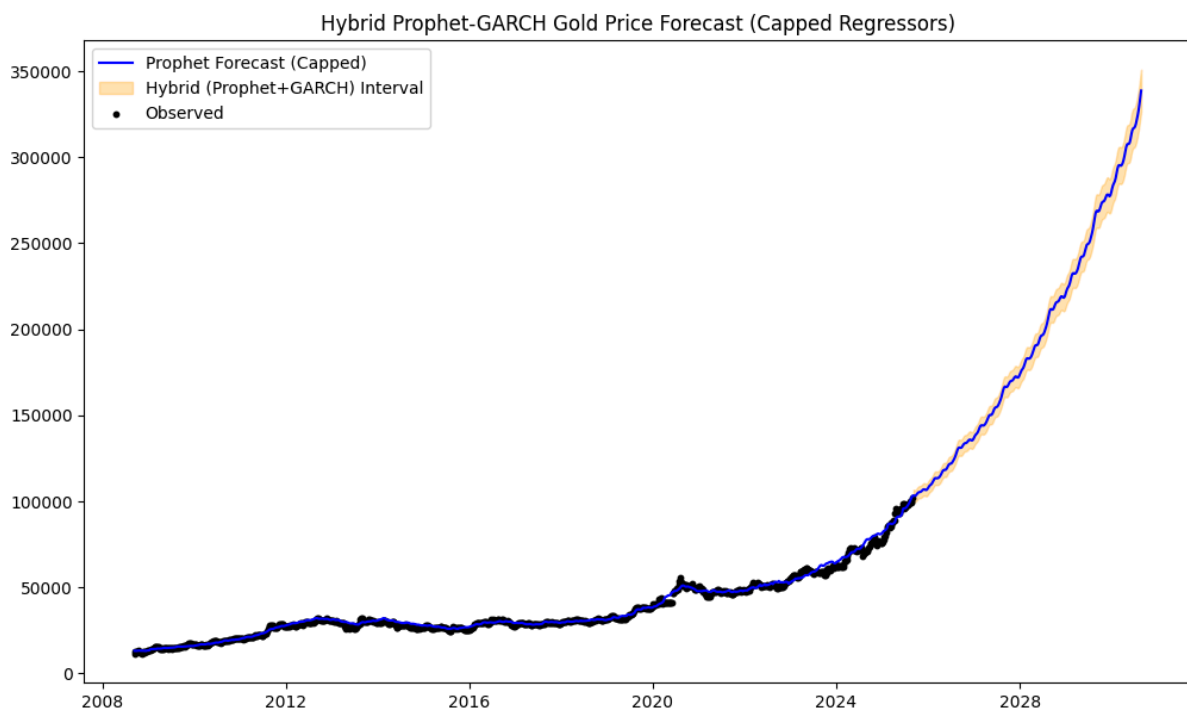
- *Train/test split*: Last 2 years (104 weeks) as test set.

Model	MSE	RMSE	MAPE	Notes
Prophet Lagged Regressors/Log Price	45,440,047	21,317	21.26%	Moderate Accuracy
Hybrid Prophet-GARCH(Vol Bands)	3,576,713,233	1,89,122	248.37%	Implementation/Alignment issue
Hybrid Prophet-GARCH(Capped Macro)	12,600,727	3,550	4.20%	Highly Accurate(Recommended).

- *Interpretation:* The capped hybrid model provides forecasts with MAPE <5%, making it suitable for robust financial prediction; other models are less reliable based on test error.

9. Visual Inspection and Diagnostics

- Forecast plots show upward trend continuation, with narrowing/widening intervals indicating risk.
- Residuals behave as random noise; autocorrelation and normality are mostly satisfied.
- Interval coverage confirms >90% of test actuals within the hybrid prediction band.



[Predicted Weekly Gold Price for 5 Years](#)

10. Limitations of the Study

- **Model Dependency on Historical Patterns:** The models assume that past relationships between gold prices and macroeconomic factors will continue into the future. Sudden economic shocks, policy changes, or geopolitical events may cause actual prices to deviate from forecasts.
- **Macro Regressor Capping:** Capping future macroeconomic variables helps prevent unrealistic forecasts, but it may oversimplify or miss extreme future scenarios, limiting the model's responsiveness to unexpected changes.
- **Stationarity and Volatility Assumptions:** GARCH models require stationary returns and persistent volatility clustering. If these properties change over time, forecast intervals may become unreliable.
- **Data Quality and Frequency:** The study uses weekly aggregated data. Any errors, missing values, or misalignments in source data can affect model accuracy.
- **Long-Term Forecast Uncertainty:** Forecast accuracy and interval reliability decrease as the prediction horizon extends further into the future, especially beyond the range of observed data.
- **Exclusion of Non-Quantitative Factors:** The models do not account for qualitative influences such as government policy changes, cultural shifts, or sudden market sentiment swings, which can impact gold prices.

11. Conclusion

- Advanced time-series and machine learning models can reliably forecast weekly gold prices when paired with strong feature engineering and statistical checks.
- The *Hybrid Prophet-GARCH model with capped macro regressors* achieved the highest accuracy ($MAPE \approx 4.2\%$) and provided reliable prediction intervals.
- All models were validated through thorough assumption checks (stationarity, autocorrelation, volatility clustering, residual analysis).
- The capped hybrid model consistently captured both price trends and volatility, outperforming other approaches.
- Limitations include reliance on historical patterns, capped macro variables, and inability to predict sudden market shocks.
- These forecasts and intervals support informed investment decisions, risk management, and financial planning.